NETFRAME

Homelab Infrastructure Runbook

km-cluster | NetFRAME CS9000 42U

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Revised:	June 22, 2026
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Contents

1 Infrastructure Overview

NetFRAME is a professional-grade homelab built inside a NetFRAME CS9000 42U rack, purpose-built as a live CCNA lab, LLM inference platform, ML research node, and backup infrastructure. The cluster is named **km-cluster** and runs Proxmox VE 9.1 across seven nodes as of June 2026.

1.1 Cluster Nodes

Hostname	Role	IP	CPU	RAM	Notes
Randy	Storage / PBS	192.168.10.187	2x E5-2690 v4	128 GB	SuperMicro CSE-219U
QuarkyLab	ML / Fernanda	192.168.10.30	2x E5-2699 v4	512 GB	R730, RTX 6000 24GB
Jarvis	LLM Inference	192.168.10.31	2x E5-2687W v4	384 GB	R730, RTX 8000 48GB
pve2	OPNsense host	192.168.10.204	i7-8700	48 GB	HP EliteDesk G4 SFF
pve3	Cluster node	192.168.10.201	i7-8700	32 GB	HP EliteDesk G4 SFF
pve4	Cluster node	192.168.10.202	i5-7500T	32 GB	HP EliteDesk G3 Mini
pve5	Cluster node	192.168.10.203	i5-7500T	32 GB	HP EliteDesk G3 Mini

1.2 Management IPs

Device	IP
EX3400 (switch)	192.168.10.50
OPNsense LAN	192.168.10.1
Randy IPMI	192.168.10.22
QuarkyLab iDRAC	192.168.10.20
Jarvis iDRAC	192.168.10.21
Pi-hole / DNS	192.168.10.177
Ares (admin)	192.168.10.100 (wired) / .199 (WiFi)

2 Network Architecture

2.1 VLAN Design

VLAN	Name	Subnet	Notes
1	mgmt	192.168.10.0/24	Management, all servers
20	trusted	192.168.20.0/24	Trusted devices
30	servers	192.168.30.0/24	Service VMs
40	IoT	192.168.40.0/24	IoT devices
50	VoIP	192.168.50.0/24	Cisco 8841 phones
60	guest	192.168.60.0/24	Guest WiFi
70	lab	192.168.70.0/24	CCNA lab segment

2.2 Switching

- **Juniper EX3400-48P** — enterprise fabric, JunOS 23.4R2-S7.4, IP 192.168.10.50

- **UniFi Switch 24 PRO (PoE+)** — consumer fabric (IoT, VoIP, guest)
- Inter-switch uplink: ge-0/0/32 (copper RJ45, active); xe-0/2/3 DAC (link-down, speed mismatch)

2.3 10G Fabric — ConnectX-3 DAC Links

Node	NIC Port	EX3400 Port
Randy (nic3)	Port 1	xe-0/2/0 — Up 10G
Randy (nic2)	Port 2	xe-0/2/2 — awaiting 4th DAC
QuarkyLab	ConnectX	xe-0/2/x
Jarvis	ConnectX	xe-0/2/x

2.4 OPNsense

OPNsense runs as VM 100 on pve2 (192.168.10.204). Version 25.7. Handles inter-VLAN routing, firewall, DHCP for all VLANs, and WAN. Serial console access: `qm terminal 100` from pve2.

3 Randy — SuperMicro CSE-219U

Randy — Commissioned June 22

[redacted]

3.1 Storage Layout

Device	Type	Purpose
/dev/sda (RAID-1)	185.8 GB SAS SSD x2	Proxmox OS boot mirror
/dev/sdb–sdu	18x 1.6TB Toshiba AL15SEB18EQ 10K SAS	ZFS datastore pool
sdc, sdg	2x 1.8TB Seagate ST2000NX0423 SATA	Spare / unallocated

3.1.1 RAID Controller — AVAGO 3108 MegaRAID

- Boot mirror: RAID-1 on slots 0:0 and 0:1 (Seagate ST200FM0053 SSDs)
- Data drives: JBOD mode enabled, all 20 data drives passed through to OS
- StorCLI installed at: `/opt/MegaRAID/storcli/storcli64`
- Symlink: `/usr/sbin/storcli64`

Listing 1: Useful StorCLI commands

```
# Show all drives and state
storcli64 /c0/eall/sall show
```

```
# Show controller info
storcli64 /c0 show

# Enable JBOD mode (if needed after reboot)
storcli64 /c0 set jbod=on
storcli64 /c0/eall/sall set jbod
```

3.1.2 ZFS Pool — datastore

Listing 2: ZFS pool layout

```
zpool status datastore
# pool: datastore
# state: ONLINE
#   raidz2-0: sdb sdd sde sdf sdh sdi
#   raidz2-1: sdj sdk sdl sdm sdn sdo
#   raidz2-2: sdp sdq sdr sds sdt sdu
# 29.4TB raw / 19.5TB usable
```

3.2 Proxmox Backup Server

- PBS version: 4.2.2
- Web UI: `https://192.168.10.187:8007`
- Datastore: `datastore` at `/datastore`
- Fingerprint: `da:61:6a:4c:49:e8:87:03:08:1d:d7:31:ab:23:58:20:47:58:e8:77:4a:52:3d:39:0c:19:52`
- Storage ID in cluster: `randy-pbs` (Shared: Yes, all nodes)

3.3 Networking

- **nic3** (Mellanox ConnectX-3 port 1): 10G UP → xe-0/2/0 on EX3400
- **nic2** (ConnectX-3 port 2): DOWN — awaiting 4th DAC cable → xe-0/2/2
- **nic0, nic1** (onboard Intel): DOWN / unused
- vmbr0 bridges nic3 for management

3.3.1 Randy Commissioning Procedure (for rebuild reference)

Listing 3: Full Randy rebuild procedure

```
# 1. Boot IPMI BIOS -- hit Ctrl+H for MegaRAID WebBIOS
#    Create RAID-1 VD on two 185GB Seagate SSDs (slots 0:0, 0:1)

# 2. Install Proxmox VE 9.1 via USB
#    Target: /dev/sda (the RAID-1 VD, 185.78GiB, SMC3108)
#    Filesystem: ext4
#    Hostname: randy.netframe.local
#    IP: 192.168.10.187/24  GW: 192.168.10.1  DNS: 192.168.10.177

# 3. Post-install repo fix
```

```
echo "# disabled" > /etc/apt/sources.list.d/pve-enterprise.list
echo "# disabled" > /etc/apt/sources.list.d/ceph.list
echo "" > /etc/apt/sources.list.d/pve-enterprise.sources
echo "" > /etc/apt/sources.list.d/ceph.sources
echo "deb http://download.proxmox.com/debian/pve bookworm pve-no-
    subscription" \
  > /etc/apt/sources.list.d/pve-no-subscription.list
apt update && apt dist-upgrade -y

# 4. Join cluster
echo "192.168.10.204 pve2.netframe.local pve2" >> /etc/hosts
pvecm add 192.168.10.204 --use_ssh

# 5. Enable JBOD passthrough
storcli64 /c0 set jbod=on
storcli64 /c0/eall/sall set jbod

# 6. Create ZFS pool
zpool create -f -o ashift=12 datastore \
  raidz2 sdb sdd sde sdf sdh sdi \
  raidz2 sdj sdk sdl sdm sdn sdo \
  raidz2 sdp sdq sdr sds sdt sdu

# 7. Install PBS
echo "deb http://download.proxmox.com/debian/pbs trixie pbs-no-
    subscription" \
  > /etc/apt/sources.list.d/pbs-no-subscription.list
apt update && apt install proxmox-backup-server -y

# 8. Create PBS datastore
proxmox-backup-manager datastore create datastore /datastore
```

4 QuarkyLab — Dell R730 #1

QuarkyLab

Role: Fernanda's ML research node
Service Tag: 1S8WR22 **iDRAC:** 192.168.10.20 (root/calvin)
CPU: Dual E5-2699 v4 (matched stepping) **RAM:** 512 GB LRDIMM ECC
GPU: RTX 6000 24 GB **BIOS:** 2.19.0

Pending: NVIDIA driver install requires kernel pinned to 6.14.11-9-pve before install (current kernel 6.17 has DRM API incompatibility with nvidia-550 driver).

Listing 4: Pin kernel before NVIDIA driver install

```
apt install pve-kernel-6.14.11-9-pve
# Set as default in /etc/default/grub, then:
update-grub && reboot
# Then install nvidia-550 driver
```

5 Jarvis — Dell R730 #2

Jarvis

[redacted]

Ollama serves Qwen2.5 72B Q4_K_M. FastAPI `llm_router.py` implements hybrid routing — local Ollama by default, escalates to Claude API when logprob confidence falls below threshold.

6 Services

6.1 Running Services

Service	Host	VM/CT ID	IP	Notes
OPNsense	pve2	VM 100	192.168.10.1	Router/firewall
Pi-hole	pve3	LXC	192.168.10.177	DNS filter
Headscale	pve3	LXC 105	192.168.10.186	VPN, v0.29.1
Wazuh	pve2	VM 104	—	SIEM
step-ca	pve2	—	—	Internal CA
PBS	Randy	—	192.168.10.187:8007	Backup server

6.2 Headscale

Headscale v0.29.1 running as LXC 105 on pve3. Ares connected as first node (Tailscale IP 100.64.0.1). Remaining devices still on commercial Tailscale pending migration.

6.3 Internal CA (step-ca)

step-ca 0.30.2 on pve2 issuing `*.netframe.local` TLS certificates.

7 Power Infrastructure

UPS	Feeds	Battery
Middle Atlantic UPS-OL2200R	R730s, Randy, DS4246	6x 12V 9Ah AGM in series (76.4V)
Tripp Lite SMART1500VA	EX3400, UniFi, small compute	Stock

PDU: APC AP7901 (ge-0/0/38 on EX3400, MAC 00:c0:b7:b4:88:4a).

8 Pending / To-Do

Order 4th DAC cable for Randy nic2 → xe-0/2/2

Pin QuarkyLab kernel to 6.14.11-9-pve, install NVIDIA 550 driver

Configure backup schedules on all nodes pointing to randy-pbs

Activate VLANs (pve2 trunk connection to EX3400 pending)

Migrate Fernanda devices from commercial Tailscale to Headscale

DS4246 connection to Randy via LSI 9207-8e HBA

FreePBX VoIP (5x Cisco CP-8841 phones acquired)

Jellyfin on Randy with RX 580 ROCm transcoding

UPS NMC IP assignment (requires USB console access)

Cyberpunk monitoring dashboard live API integration

RKE2 / Kubernetes planning (Cilium, MetalLB, NVIDIA GPU Operator)

9 Quick Reference Commands

9.1 Cluster Health

```
# From any node
pvecm status
pvecm nodes

# PBS status
systemctl status proxmox-backup
proxmox-backup-manager datastore list

# ZFS pool health
zpool status datastore
zpool list
```

9.2 EX3400 Switch

```
ssh mason@192.168.10.50
# Always exit to > operational mode before show commands
# Paste one line at a time

show interfaces xe-0/2/0 brief   # Randy 10G port 1
show interfaces xe-0/2/2 brief   # Randy 10G port 2
show ethernet-switching table     # MAC table
show lldp neighbors               # Connected devices
```

9.3 Randy Storage

```
ssh root@192.168.10.187

# Check all drives
storcli64 /c0/eall/sall show

# ZFS status
zpool status datastore
```

```
zfs list

# If drives go dark after reboot (JBOD mode reset)
storcli64 /c0 set jbod=on
storcli64 /c0/eall/sall set jbod
```